

VoxNeuro: Leakage-Safe Parkinson’s Screening Support via Multi-View Grassmannian Speech Analysis

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Abstract: Parkinson’s disease (PD) alters phonation and articulation, making repeated speech a candidate digital biomarker. Public speech datasets contain several recordings per participant, and record-wise cross-validation can place one person in both training and test sets, so reported scores partly reflect recognition of an observed person rather than generalization to unseen participants. VoxNeuro treats the subject as the observation: each subject’s three recordings define a nominal rank-three Grassmann subspace and a Euclidean mean-dispersion vector, whose Gaussian kernels are fused into a positive-semidefinite kernel for support vector classification. Two fold-local components address heavy-tailed descriptors and high-dimensional distance concentration: Gaussian rank normalization and, when features outnumber training recordings, a supervised-subspace ensemble built in low-dimensional partial-least-squares spaces. At the seed-42 development partition of the subject-disjoint protocol, the adaptive model reached balanced accuracies of 0.875 (UCI-489) and 0.816 (PD-252), with PD-252 sensitivity 0.899 and specificity 0.734. In 20 further overlapping partitions of the same subjects, its PD-252 estimates exceeded the original model’s in every partition (+0.062 balanced accuracy, +0.050 macro-F1), whereas UCI-489 differences were about +0.001; most of the gain came from normalization and the supervised Euclidean summary rather than the Grassmann term. These analyses assess partition sensitivity and are not independent validation.

Keywords: Parkinson’s disease; speech biomarker; Grassmann manifold; kernel fusion; imbalanced learning; subject-disjoint cross-validation; support vector machine; rank normalization; partial least squares; digital health

1. Introduction

Parkinson’s disease (PD) is a progressive neurodegenerative disorder involving nigrostriatal dopaminergic loss, basal ganglia circuit dysfunction, and heterogeneous motor and non-motor manifestations [1–3]. Its diagnosis remains clinical [4,5], and its global burden has grown faster than that of any other neurological disorder [6,7]. Disordered speech is among the most common motor consequences: dysarthria, hypophonia, and related vocal-tract dysfunctions affect the large majority of patients over the disease course [8,9]. Quantitative acoustic analysis detects these changes even in early and untreated disease [10]. Nonlinear dysphonia measures separate disordered from healthy voices [11], and repeated noninvasive speech tests have supported research on low-burden remote monitoring and progression tracking [12–14]. Automated speech analysis has revealed prodromal changes in at-risk cohorts [15]. Machine-learning approaches to voice-based PD detection have accordingly expanded rapidly, as documented by recent systematic reviews [16,17]. A digital biomarker, however, is clinically credible only when its evaluation matches its intended unit of use: the patient.

Public PD speech datasets often contain multiple recordings per participant. When recordings are shuffled independently, subject-specific anatomical and acquisition signatures can occur on both sides of a record-wise split, and the evaluation then estimates recognition of another recording from a previously observed person rather than generalization to an unseen person. This failure mode is a specific instance of data leakage, a recognized driver of over-optimistic results across applied machine learning [18,19]. This effect has been demonstrated directly in clinical prediction: record-wise cross-validation can inflate accuracy substantially relative to subject-wise evaluation of the very same pipeline [20], a finding that prompted an extended methodological discussion [21]. Subject characteristics can also drive apparent performance even when splits look innocuous [22]. Related biases arise whenever

model selection and error estimation share data [23,24], and cross-validation on small samples carries wide error bars even under favorable conditions [25–27]. The UCI-489 corpus analyzed here contains three dependent replications from each of 80 subjects [28]; replication-aware analysis has been proposed for exactly this cohort [29].

Methodologically, three research strands converge in this study. First, engineered speech descriptors: dysphonia, cepstral, wavelet, and tunable Q-factor wavelet transform (TQWT) descriptor families have been used to characterize PD-related voice changes [12,30,31]. Second, subspace geometry: collections of related observations can be represented as points on the Grassmann manifold $\text{Gr}(r, d)$, whose quotient structure removes arbitrary basis choices [32–34]; this representation underpins subspace-based learning in computer vision and pattern recognition [35,36], and kernels on such manifolds require care because Gaussian functions of arbitrary manifold distances are not automatically positive semidefinite (PSD) [37]. Third, kernel fusion: reproducing-kernel Hilbert space (RKHS) theory [38–40] and multiple-kernel learning [41–43] combine heterogeneous similarities while preserving convex support vector optimization. Class imbalance, finally, is routinely addressed by synthetic oversampling [44–46], whose behavior in high-dimensional feature spaces warrants caution [47]. This study combines these strands under an evaluation protocol that treats the patient as the statistical unit.

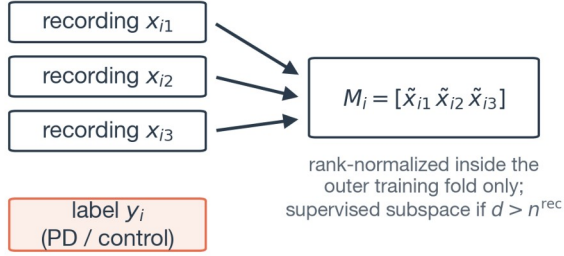
Deep architectures, including convolutional, recurrent, and transformer-based models, are increasingly applied to voice-based PD detection and scale well when large corpora are available [16]. Smartphone-scale collection efforts such as mPower point in that direction [48,49], and technology roadmaps for PD anticipate exactly this kind of remote, high-frequency assessment [50–52]. The cohorts relevant to replicated-phonation screening, however, typically contain tens to a few hundred subjects. In this regime, the present study deliberately combines structured geometric priors with a convex, auditable classical learner instead of high-capacity end-to-end models: each component (representation, kernel, solver, and resampling step) has an explicit mathematical description that can be verified directly against its implementation.

VoxNeuro aligns the mathematical unit of analysis with the clinical unit of interest: each subject is represented by the complete set of repeated measurements. A fixed-rank representation based on singular value decomposition (SVD) encodes the joint orientation of the repetitions, while a Euclidean summary preserves feature-space location and within-subject variability. A positive-semidefinite direct-sum kernel combines the two views, and evaluation is subject-disjoint throughout (Figure 1). For imbalanced data, synthetic subjects are generated along manifold geodesics rather than by interpolating individual recording rows. The study had three objectives: (i) to formalize why record-wise and subject-disjoint evaluation estimate different risks; (ii) to assess whether adding the Grassmann kernel to a matched nonlinear Euclidean model improves subject-level classification; and (iii) to test whether two fold-local remedies for heavy-tailed descriptors and high-dimensional distance concentration, Gaussian rank normalization and a supervised-subspace stage, improve subject-level classification, with all fitted transformations restricted to each outer training fold; the adaptive specification itself was developed after reviewing seed-42 cross-validation results.

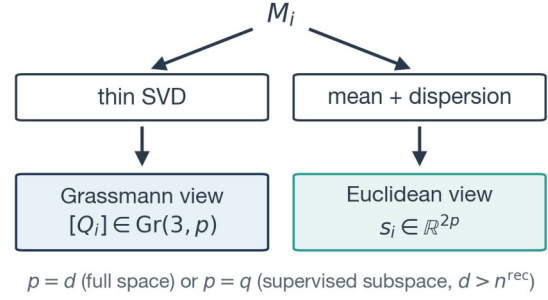
The principal contributions are as follows: (i) a formal subject-disjoint risk definition together with a quantitative account of the optimistic bias induced by record-wise splitting; (ii) a basis-invariant Grassmann representation of repeated speech; (iii) provably positive-semidefinite fusion of geometric and Euclidean kernels; (iv) geodesic minority oversampling coupled to the Euclidean view; (v) a rank-normalized, dimension-adaptive extension whose supervised-subspace stage is activated only when the feature count exceeds the number of training recordings, evaluated with an explicit distinction between the development partition and an internal resampling-sensitivity analysis over 20 further partitions; and (vi) a separate research-stage web application for guided speech capture and longitudinal result visualization, not used in the present analyses. One fixed protocol is applied to both cohorts with a common comparator set, including a matched nonlinear Euclidean ablation with a radial basis function (RBF) kernel that isolates the contribution of the Grassmann kernel term, and robustness

to the augmentation seed, the outer subject partition, the subspace rank, the augmentation family, the subspace dimension, and the normalization is reported.

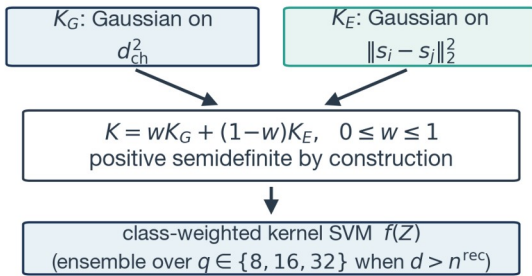
A Subject-level observation



B Two complementary views



C PSD kernel fusion



D Leakage-safe evaluation

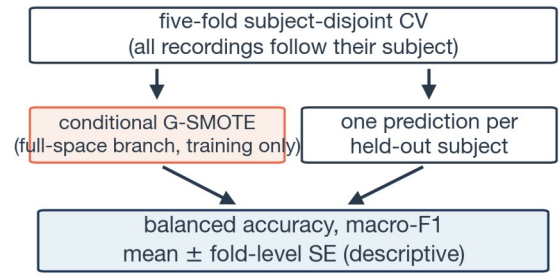


Figure 1. Schematic overview of the VoxNeuro framework. (A) Each subject contributes three repeated recordings and one diagnostic label; the recordings are normalized with functions fitted inside the outer training fold only (Gaussian rank normalization) and stacked into a matrix M_i ; when the feature count exceeds the number of training recordings, the recordings are first projected onto a supervised subspace (Section 2.10). (B) A thin singular value decomposition (SVD) yields the Grassmann view $[Q_i]$, while coordinate-wise mean and dispersion yield the Euclidean view s_i ; the views live in $\mathbb{R}^p$ with $p=d$ in the full space and $p=q$ in the supervised subspace. (C) Gaussian kernels on the two views are combined into a positive-semidefinite (PSD) fused kernel feeding a class-weighted support vector machine (SVM). (D) Evaluation is five-fold subject-disjoint cross-validation (CV); in the full-space branch, conditional Grassmann-geodesic synthetic minority oversampling (G-SMOTE) is activated only inside imbalanced training folds, and fold scores are summarized as mean $\pm$ standard error (SE). PD, Parkinson’s disease.

2. Materials and Methods

2.1. Datasets

Two public, de-identified cohorts were analyzed (Table 1). The first, denoted UCI-489, comprises 80 subjects (40 PD, 40 control) with three recordings of sustained /a/ per subject, each recording described by 44 replicated acoustic descriptors and one dataset-provided binary sex/gender variable [28,29]. The second, denoted PD-252 [53], comprises 252 subjects (188 PD, 64 control) with three recordings per subject, described by 752 descriptors spanning baseline acoustic, vocal/time-frequency, Mel-frequency cepstral coefficient (MFCC) [54], wavelet, TQWT [55], and intrinsic mode function/empirical mode decomposition (IMF/EMD) [56] families [30,31]. It also includes one dataset-provided binary sex/gender variable, which constitutes the demographic family of the permutation analysis in Section 3.4. The coded variable was included as a fixed model input and was never used to construct folds. The cohorts were selected as complementary evaluation regimes for the same repeated-measurement protocol: UCI-489 evaluates the representation in a balanced, lower-dimensional setting without

synthetic augmentation, whereas PD-252 evaluates the same representation in a larger, high-dimensional, imbalanced setting in which the dimensionality gate of Section 2.10 activates the supervised-subspace stage and the full-space comparators exercise the prespecified imbalance-handling branch. This study was a secondary computational analysis of public data and involved no participant recruitment.

Table 1. Dataset regimes and evaluation design.

Attribute	UCI-489	PD-252
Subjects	80	252
PD / control	40 / 40	188 / 64
Recordings per subject	3	3
Modeled variables	45	753
Acoustic regime	Sustained / a/	Multi-family engineered features
Imbalance handling	Class weighting (balanced folds)	Class weighting; G-SMOTE gate in the full-space branch only
Outer evaluation	Five-fold subject-disjoint cross-validation	

2.2. Subject-Disjoint Problem Formulation

Let $R_i = \{(x_{ij}, y_i) : j = 1, \dots, m_i\}$ denote all recordings from subject i ; Table 2 summarizes the notation used throughout. For outer fold k , let I_k^{tr} and I_k^{te} be the training and held-out subject-index sets. Leakage-safe evaluation requires

$$I_k^{\text{tr}} \cap I_k^{\text{te}} = \emptyset, I_k^{\text{tr}} \cup I_k^{\text{te}} = \{1, \dots, n\}. \quad (1)$$

All recordings in R_i are assigned to the same partition as subject i (Figure 2A,B). For a pointwise classification loss ℓ , the fold risk is evaluated once per held-out subject,

$$\hat{R}_k(f) = \frac{1}{|I_k^{\text{te}}|} \sum_{i \in I_k^{\text{te}}} \ell(f(Z_i), y_i), \quad (2)$$

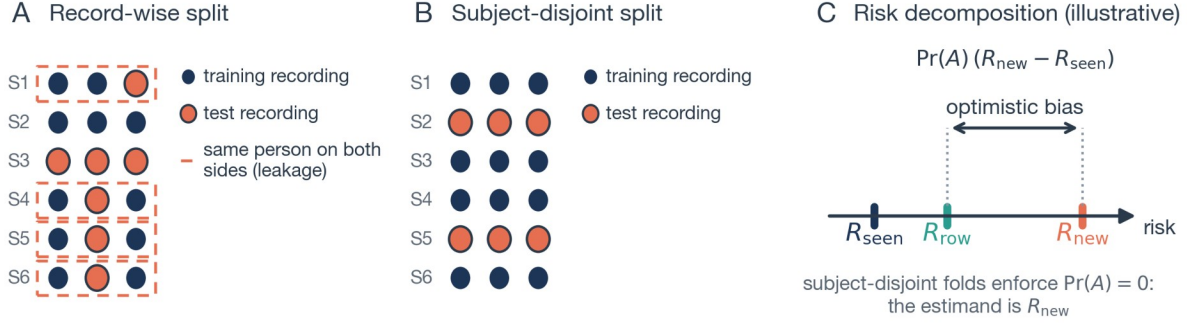
where Z_i is the complete subject-level representation. Equations (1) and (2) define the intended estimand, generalization to an unseen person rather than to an unseen row from a known person; the reported balanced accuracy and macro-F1 are fold-level aggregate metrics defined in Equation (15).

Table 2. Principal notation.

Symbol	Meaning
$x_{ij} \in \mathbb{R}^d$	feature vector of recording j of subject i
R_i, y_i	all recordings of subject i and its diagnostic label
$I_k^{\text{tr}}, I_k^{\text{te}}$	training and held-out subject-index sets of outer fold k
M_i	standardized $d \times 3$ recording matrix of subject i
$Q_i, [Q_i]$	orthonormal basis from the thin SVD and its subspace class
$\text{Gr}(r, d), \text{St}(r, d)$	Grassmann and Stiefel manifolds of rank r in $\mathbb{R}^d$
$P_i = Q_i Q_i^T$	orthogonal projector representing $[Q_i]$
$\theta_{ij, q'}, d_{\text{ch}}$	principal angles and chordal distance between subspaces
$s_i \in \mathbb{R}^{2d}$	Euclidean mean–dispersion summary of subject i
$K_G, K_E, K; w$	Grassmann, Euclidean, and fused kernels; fusion weight
γ_G, γ_E	Gaussian kernel coefficients (inverse squared bandwidths) of the two views
$Q_{\text{syn}}(t), s_{\text{syn}}(t)$	geodesic and Euclidean interpolants of G-SMOTE
C_{y_i}	class-weighted SVM box constraint for class y_i
$F, \bar{m}, \text{SE}$	number of outer folds, fold-mean metric, standard error

$\hat{I}_{gk'} \pi_{g'} B$	fold-level permutation importance (mean over B draws) and permutation of family g
$h, A, R_{\text{seen}}, R_{\text{new}}, R_{\text{row}}$	record-level classifier, leakage event, and associated risks
$\hat{F}_k^{(l)}, \Phi^{-1}$	training-fold empirical distribution function of coordinate l ; standard normal quantile function
$W_{q'} \tau_{ij'} Q$	PLS rotation to q dimensions, class-balanced recording targets, set of subspace dimensions
$f_{AS'} \hat{\sigma}_{k,q'} n_k^{\text{rec}}$	adaptive-subspace decision function, training-fold decision scale of member q , number of training recordings

142



143

144 **Figure 2.** Schematic illustration of evaluation leakage with repeated recordings. (A) A record-wise split assigns
145 individual recordings to training or test sets, so several subjects (dashed outlines) appear on both sides and the test
146 set is not independent of training identities. (B) A subject-disjoint split assigns all recordings of a subject jointly. (C)
147 Consequence of Equation (18): the record-wise estimate R_{row} interpolates between the easier seen-person risk R_{seen}
148 and the deployment-relevant unseen-person risk R_{new} , drawn for an illustrative $\Pr(A) = 0.65$; the arrow marks the
149 optimistic bias of magnitude $\Pr(A)(R_{\text{new}} - R_{\text{seen}})$.

150 2.3. Standardization and Subject Representation

151 For subject i , the three acoustic vectors $x_{ij} \in \mathbb{R}^d$ are standardized using a mean vector μ_k and a scale vector
152 estimated exclusively from the recordings of training subjects in outer fold k . Under the implemented
153 standardization procedure, a zero-variance training coordinate is assigned unit scale:

$$154 \quad \begin{aligned} \tilde{x}_{ij} &= D_k^{-1}(x_{ij} - \mu_k), D_k = \text{diag}(\sigma_k'), \\ M_i &= [\tilde{x}_{i1} \tilde{x}_{i2} \tilde{x}_{i3}], \end{aligned} \quad (3)$$

155 where σ_k' denotes the training-fold scale vector after zero-variance handling. A thin SVD [57] gives

$$156 \quad \begin{aligned} M_i &= U_i \Sigma_i V_i^T, Q_i = U_i(:, 1:r), \\ Q_i^T Q_i &= I_r, [Q_i] \in \text{Gr}(r, d). \end{aligned} \quad (4)$$

157 Throughout, the rank is fixed at $r=3$, matching the three repeated recordings per subject. The basis is
158 not unique: Q_i and $Q_i O$, with O in the orthogonal group $O(r)$, represent the same subspace. This
159 quotient structure removes coordinate choices within the retained SVD basis (Figure 3A). In the official
160 releases, one subject per cohort (CONT-36 in UCI-489; subject 37 in PD-252) contains byte-identical
161 repeated recordings, so its matrix has numerical rank two. For such a subject, Equation (4) retains the
162 orthonormal completion supplied by the SVD as the third basis vector; that direction is a solver-
163 dependent completion that is not identified by the recordings, so the representation is nominally rank
164 three for these two subjects, and the reference implementation accepts them only under an explicit opt-
165 in flag. Equation (3) defines the original configuration; the adaptive configuration of Section 2.10
166 replaces the standardization by Gaussian rank normalization and, in the high-dimensional regime,
167 builds M_i from supervised low-dimensional projections of the recordings.

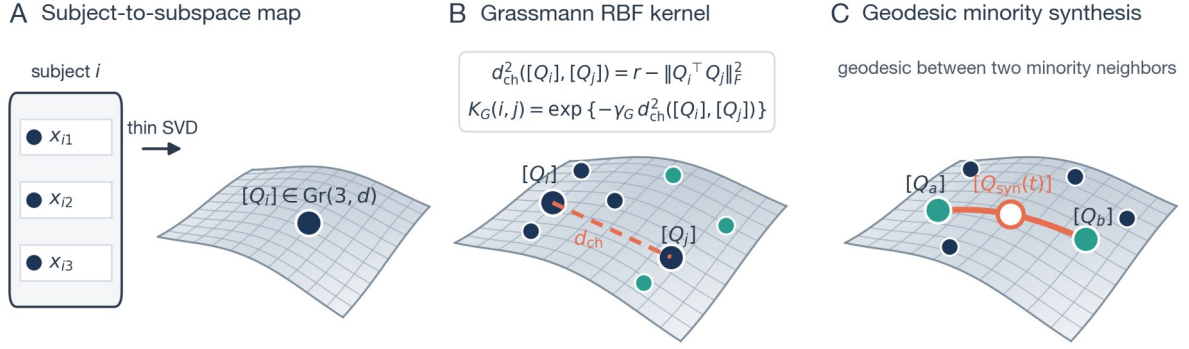


Figure 3. Geometric interpretation of the VoxNeuro representation (schematic; the curved sheet stands for the Grassmann manifold). (A) The three repeated recordings of a subject are mapped by a thin SVD to a single point $[Q_i]$ on $\text{Gr}(3, d)$. (B) The chordal distance d_{ch} between two subject subspaces (dashed chord), whose square enters the displayed Gaussian Grassmann radial-basis-function kernel, compares subjects independently of the arbitrary orthonormal basis selected for each subspace; navy and teal points denote subjects of the two classes. (C) Schematic of minority synthesis along the Grassmann geodesic joining two minority neighbors, rather than row-wise Euclidean interpolation.

2.4. Principal Angles and Projection Geometry

Principal angles between two subject subspaces (Figure 4A) are defined by [32,58]

$$Q_i^\top Q_j = A_{ij} \cos(\Theta_{ij}) B_{ij}^\top, \quad \Theta_{ij} = \text{diag}(\theta_{ij,1}, \dots, \theta_{ij,r}), \quad (5)$$

where A_{ij} and B_{ij} are orthogonal. With the projector $P_i = Q_i Q_i^\top$, the squared chordal distance is given by [33,59]

$$\begin{aligned} d_{\text{ch}}^2([Q_i], [Q_j]) &= \frac{1}{2} \|P_i - P_j\|_F^2 \\ &= r - \|Q_i^\top Q_j\|_F^2 = \sum_{q=1}^r \sin^2 \theta_{ij,q}. \end{aligned} \quad (6)$$

Because $(Q_i O)(Q_i O)^\top = P_i$, the distance is basis invariant (Figure 4B). Vectorizing P_i maps the chordal distance to a Euclidean distance in projector space, up to a constant factor $\sqrt{2}$ absorbed by the kernel coefficient, thereby providing a Euclidean embedding for the Gaussian Grassmann kernel.

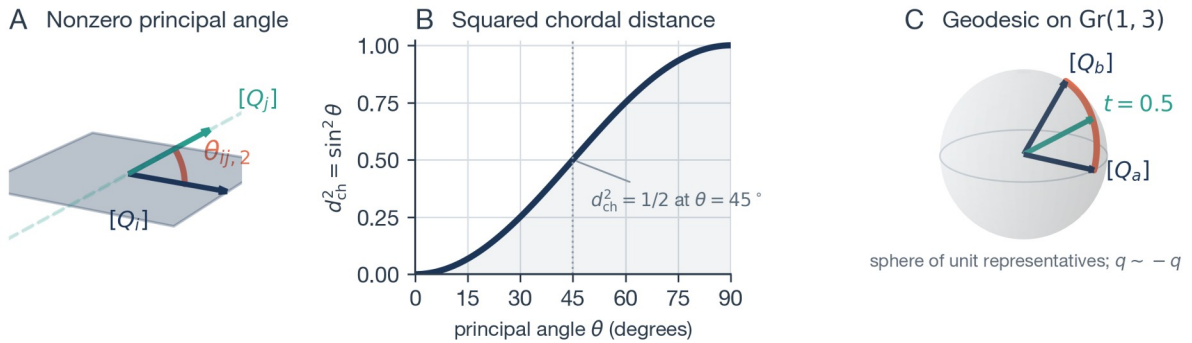


Figure 4. The geometry behind Equations (5), (6), (11), and (12), drawn for illustrative toy subspaces. (A) Two planes in $\mathbb{R}^3$ that share one direction, so the first principal angle is zero and the drawn angle is the second principal angle $\theta_{ij,2}$. (B) Exact rank-one relation between a principal angle and the squared chordal distance, $d_{\text{ch}}^2 = \sin^2 \theta$. (C) A geodesic path computed with Equations (11) and (12) between two one-dimensional subspaces $[Q_a]$ and $[Q_b]$; every interpolated frame in the drawn path satisfies $Q_{\text{syn}}(t)^\top Q_{\text{syn}}(t) = I_r$ to machine precision, consistent with Proposition 2.

2.5. Complementary Euclidean View

The complementary Euclidean view retains the coordinate-wise mean and dispersion:

$$\begin{aligned} \mu_i &= \frac{1}{3} \sum_{j=1}^3 \tilde{x}_{ij}, \sigma_i = \left[\frac{1}{3} \sum_{j=1}^3 (\tilde{x}_{ij} - \mu_i)^2 \right]^{1/2}, \\ s_i &= [\mu_i^T \sigma_i^T]^T \in \mathbb{R}^{2d}, \end{aligned} \quad (7)$$

where $(\cdot)^{\odot 2}$ and the square root are applied elementwise. Thus $[Q_i]$ encodes the orientation of the span of the standardized repetitions (M_i is not centered within subject, so this span generally includes the mean direction), whereas s_i retains coordinate-level location and dispersion.

2.6. Direct-Sum Kernel and Guarantee of Positive Semidefiniteness

Let ϕ_G and ϕ_E be feature maps for Gaussian kernels on the projection and Euclidean views [38,39]. With fusion weight w , the joint representation is the Hilbert direct sum

$$\Phi(Z_i) = (\sqrt{w} \phi_G([Q_i]), \sqrt{1-w} \phi_E(s_i)), \quad 0 \leq w \leq 1, \quad (8)$$

which induces, for kernel coefficients $\gamma_G, \gamma_E > 0$ (inverse squared bandwidths),

$$\begin{aligned} K_G(i, j) &= \exp\{-\gamma_G d_{\text{ch}}^2([Q_i], [Q_j])\}, \\ K_E(i, j) &= \exp\{-\gamma_E \|s_i - s_j\|_2^2\}, \\ K(i, j) &= w K_G(i, j) + (1-w) K_E(i, j). \end{aligned} \quad (9)$$

Proposition 1 (*basis invariance and positive semidefiniteness*). The kernel in Equation (9) is invariant to $Q_i \mapsto Q_i O$ and is PSD. Basis invariance follows from the projector representation in Equation (6). Each Gaussian block is PSD on its Euclidean embedding, and a nonnegative weighted sum of PSD kernels is PSD [39,41]; hence, for every $c \in \mathbb{R}^n$,

$$c^T K c = w c^T K_G c + (1-w) c^T K_E c \geq 0. \quad (10)$$

Equivalently, $K(i, j) = \langle \Phi(Z_i), \Phi(Z_j) \rangle$ in the direct-sum reproducing-kernel Hilbert space. $\square$ The projector embedding is essential: Gaussian kernels built on other manifold distances, such as the geodesic arc length, are not PSD for all bandwidths [37], whereas the chordal construction used here inherits positive semidefiniteness from its explicit Euclidean embedding.

2.7. Grassmann-Geodesic Synthetic Minority Oversampling

This project-specific manifold adaptation is referred to as Grassmann-geodesic SMOTE (G-SMOTE); the abbreviation does not denote the separate Geometric SMOTE algorithm of Euclidean imbalanced learning [60]. Following the synthetic-neighbor principle of standard SMOTE [44] and its descendants [45,46,61], for minority subjects a and b ,

$$\begin{aligned} Q_a^T Q_b &= U \cos(\Theta) V^T, \\ W_{ab} &= (Q_b V - Q_a U \cos \Theta)(\sin \Theta)^\dagger, \end{aligned} \quad (11)$$

where $\dagger$ denotes the Moore–Penrose pseudoinverse. A shortest geodesic and its coupled Euclidean interpolation are defined as follows (Figure 4C):

$$\begin{aligned} Q_{\text{syn}}(t) &= [Q_a U \cos(t\Theta) + W_{ab} \sin(t\Theta)] V^T, \\ s_{\text{syn}}(t) &= s_a + t(s_b - s_a), \quad t \sim U(0, 1). \end{aligned} \quad (12)$$

Within each imbalanced training fold, the minority class was the control class. For each synthetic subject, one minority subject was sampled uniformly at random, and a partner was selected uniformly from that subject's five nearest minority neighbors under squared chordal distance. Synthetic subjects were generated until the two class counts were equal. The same sampled value of t was used for the Grassmann and Euclidean interpolations. For numerical stabilization, the implementation applied a QR factorization to $Q_{\text{syn}}(t)$. The resulting invertible right basis change preserves the represented Grassmann point. No synthetic subjects were generated for balanced UCI-489. Interpolating whole-subject representations along the manifold avoids creating synthetic recording rows and selects partners by subspace geometry, but it still interpolates high-dimensional summaries and therefore remains susceptible to the noise-amplification risks of oversampling [47].

Proposition 2 (*manifold validity*). In exact arithmetic, the nonzero-angle columns of $Q_a U$ and W_{ab} are mutually orthonormal, while zero-angle directions remain in the cosine term. Consequently,

$$Q_{\text{syn}}(t)^\top Q_{\text{syn}}(t) = V [\cos^2(t\Theta) + \sin^2(t\Theta)] V^\top = I_r. \quad (13)$$

Therefore, $Q_{\text{syn}}(t) \in \text{St}(r, d)$, the Stiefel manifold of orthonormal $d \times r$ frames; it represents a valid Grassmann point. Its endpoints are equivalent to $[Q_a]$ and $[Q_b]$. $\square$

2.8. Kernel Support Vector Machine, Metrics, and Complexity

For support vector machine (SVM) training [62], labels are recoded as $y_i \in \{-1, +1\}$, $Y = \text{diag}(y_1, \dots, y_n)$, and balanced class weighting sets $C_{y_i} = Cn / (2n_{y_i})$, where n_{y_i} is the number of training subjects in class y_i ; this is the balanced heuristic implemented in standard open-source libraries [63]. When the training classes are exactly balanced, as on UCI-489 and after G-SMOTE parity on PD-252, these weights reduce to $C_{y_i} = C$; the weighting is therefore active for the unaugmented class-weighted comparators and for the supervised-subspace branch of the adaptive model, which generates no synthetic subjects, on imbalanced PD-252. The class-weighted dual and the decision function with bias b are

$$\begin{aligned} \max_{\alpha} \quad & 1^\top \alpha - \frac{1}{2} \alpha^\top Y K Y \alpha, \\ \text{subject to} \quad & 0 \leq \alpha_i \leq C_{y_i}, y^\top \alpha = 0, \\ f(Z) \quad & = \sum_{i=1}^n \alpha_i y_i K(Z_i, Z) + b. \end{aligned} \quad (14)$$

Balanced accuracy (BA) [64] and macro-F1 [65] weight diagnostic classes symmetrically:

$$\begin{aligned} \text{BA} &= \frac{1}{2} (\text{TPR} + \text{TNR}), \\ \text{macro-F1} &= \frac{1}{2} (F1_{\text{PD}} + F1_{\text{Ctrl}}), \end{aligned} \quad (15)$$

where TPR is sensitivity, TNR is specificity, and $F1_{\text{PD}}$ and $F1_{\text{Ctrl}}$ are the class-wise F1 scores. The Matthews correlation coefficient is a common alternative [66]; balanced accuracy and macro-F1 were retained because both weight the two classes symmetrically and are directly interpretable as class-wise rates. For $F=5$ outer folds, a metric m is summarized by

$$\begin{aligned} \bar{m} &= \frac{1}{F} \sum_{k=1}^F m_k, \\ \text{SE}(\bar{m}) &= \left[\frac{1}{F(F-1)} \sum_{k=1}^F (m_k - \bar{m})^2 \right]^{1/2}. \end{aligned} \quad (16)$$

The reported standard error (SE) is a descriptive measure of fold-to-fold variability and is not an inferential confidence interval.

For n subjects and rank $r=3$, subject SVDs require $O(n d r^2)$, the Grassmann Gram matrix requires $O(n^2 d r^2)$, and the Euclidean block requires $O(n^2 d)$. These bounds exclude the kernel-SVM optimization and the additional G-SMOTE neighbor search, which can dominate as n grows; all components remain tractable for the present cohort sizes $n=80$ and $n=252$.

2.9. Formal Effect of Record-Wise Leakage

The mechanism illustrated in Figure 2 can be stated formally, following record-wise versus subject-wise analyses in clinical machine learning [18,20,21]. Let (X, y) be a random validation recording and its label, let A be the event that the recording belongs to a person represented in training, and let h be a record-level classifier. With A^c the complement of A , define

$$\begin{aligned} R_{\text{seen}} &= \mathbb{E}[\ell(h(X), y) | A], \\ R_{\text{new}} &= \mathbb{E}[\ell(h(X), y) | A^c]. \end{aligned} \quad (17)$$

A record-wise split then estimates

$$\begin{aligned} R_{\text{row}} &= \Pr(A) R_{\text{seen}} + [1 - \Pr(A)] R_{\text{new}}, \\ R_{\text{row}} - R_{\text{new}} &= \Pr(A) (R_{\text{seen}} - R_{\text{new}}). \end{aligned} \quad (18)$$

If repeated recordings from previously observed people are easier to classify than recordings from unseen people, then $R_{\text{seen}} < R_{\text{new}}$, and the record-wise estimate is optimistically biased in direct proportion to $\Pr(A)$ (Figure 2C). Subject-disjoint folds enforce $\Pr(A) = 0$ by construction.

2.10. Rank Normalization and the Dimension-Adaptive Supervised Subspace

Two fold-local components were added to the original configuration after inspection of the default-partition results (Section 2.13 states the resulting development-versus-confirmation distinction). The first replaces the z-standardization of Equation (3) by Gaussian rank normalization. For coordinate l , let $\hat{F}_k^{(l)}$ denote the empirical distribution function of the training-fold recordings of outer fold k , approximated through $J_k = \min(300, n_k^{\text{rec}})$ empirical quantiles with linear interpolation between them ($J_k = 192$ on UCI-489 and 300 on PD-252, where n_k^{rec} is the number of training recordings), and let Φ^{-1} denote the standard normal quantile function, with its argument clipped to $[10^{-7}, 1 - 10^{-7}]$:

$$\tilde{x}_{ij}^{(l)} = \Phi^{-1}\left(\hat{F}_k^{(l)}(x_{ij}^{(l)})\right), \quad l = 1, \dots, d, \quad (19)$$

The transformation is monotone in every coordinate and is fitted on training recordings only; held-out recordings are mapped through the frozen training-fold functions. Many engineered acoustic descriptors are heavy-tailed, and under z-standardization a few extreme coordinates dominate the squared Euclidean distances that enter K_E and the median bandwidth rule. Rank normalization equalizes the marginal scale of all coordinates without any use of labels.

The second component addresses distance concentration. When the feature count d is large relative to the number of recordings, the rank-three subspaces of different subjects become nearly orthogonal, and the squared chordal distances of Equation (6) crowd toward their maximum value r ; a Gaussian kernel on nearly constant distances carries little information. On PD-252 the coefficient of variation of the pairwise chordal distance between training subjects (means over the five seed-42 training folds) was 0.019 under z-standardization and 0.035 under rank normalization, compared with 0.052 and 0.066 on the 45-dimensional UCI-489 cohort. The proposed remedy is a supervised low-dimensional projection fitted on the training-fold recordings: a partial least squares (PLS) regression [67,68] of the normalized recordings on class-balanced targets $\tau_{ij} = y_i / n_{k,y}^{\text{rec}}$, where $y_i \in \{-1, +1\}$ is the label of subject i , carried by each of its recordings, and $n_{k,y}^{\text{rec}}$ is the number of training recordings with label y . The regression was fitted without target or predictor scaling; with $W_q \in \mathbb{R}^{d \times q}$ the fitted rotation matrix and $\hat{\tilde{x}}_k$ the mean of the normalized training recordings, every recording is projected as

$$\begin{aligned} z_{ij} &= W_q^T (\tilde{x}_{ij} - \hat{\tilde{x}}_k) \in \mathbb{R}^q, \tau_{ij} = \frac{y_i}{n_{k,y}^{\text{rec}}}, \\ M_i^{(q)} &= [z_{i1} z_{i2} z_{i3}], [Q_i^{(q)}] \in \text{Gr}(3, q), s_i^{(q)} \in \mathbb{R}^{2q}, \end{aligned} \quad (20)$$

and the subject views of Sections 2.3–2.5, the median bandwidth rule, and the fused kernel of Equation (9) are then built in $\mathbb{R}^q$ exactly as in the full space, with the prespecified weight $w = 0.5$. Because the projection is supervised, it is refitted inside every training fold and never sees a held-out recording. After projection to $q = 16$ dimensions the coefficient of variation of the PD-252 chordal distances rose to 0.064. One class-weighted SVM f_q is trained for each dimension $q \in Q = \{8, 16, 32\}$; instead of selecting a dimension within the fold, the decision values of the three members are divided by their training-fold standard deviations $\hat{\sigma}_{k,q}$ (population standard deviation of the members' decision values over the training subjects, plus 10^{-12}) and averaged, giving a fixed-dimension ensemble that performs no within-fold dimension selection:

$$\begin{aligned} f_{\text{AS}}(Z) &= \frac{1}{|Q|} \sum_{q \in Q} \frac{f_q(Z)}{\hat{\sigma}_{k,q}}, \hat{\sigma}_{k,q} = \left[\frac{1}{|I_k^{\text{tr}}|} \sum_{i \in I_k^{\text{tr}}} (f_q(Z_i) - \bar{f}_{k,q})^2 \right]^{1/2} + 10^{-12}, \\ \text{activated iff } d &> n_k^{\text{rec}} = \sum_{i \in I_k^{\text{tr}}} m_i. \end{aligned} \quad (21)$$

The stage is activated by a fixed dimensionality gate: it is applied only when d exceeds the number of training recordings n_k^{rec} , which holds on PD-252 ($d = 753$, $n_k^{\text{rec}} = 603$ or 606) and not on UCI-489 ($d = 45$,

$n_k^{\text{rec}} = 192$). The gate, the dimension set, and the equal-weight average were fixed before the partition analysis of Section 2.13 but after both cohorts had been inspected at the development partition. When the gate is inactive the model reduces to the original fused model of Equation (9) under rank normalization, including the conditional G-SMOTE branch of Section 2.7. Inside the supervised subspace no synthetic subjects are generated: the class-weighted SVM of Section 2.8 balances the classes, and Section 3.5 reports that geodesic oversampling in the compressed space lowered balanced accuracy. The complete adaptive model is referred to as the adaptive-subspace fused model; its full-space branch under rank normalization is referred to as the rank-normalized fused model, and the z-standardized configuration of the original submission as the original fused model. All three share the folds, the representation, the kernel family, the fusion weight, and the SVM penalty.

2.11. Experimental Protocol

Outer folds were generated by five-fold stratified subject-level cross-validation with shuffling and a random seed of 42 (Figure 5A). For each outer fold, the recording-level normalizer (z-standardization for the original configuration, Gaussian rank normalization for the adaptive model) was fitted only to recordings from training subjects. Each kernel coefficient was estimated from the positive squared pairwise distances δ^2 of the corresponding view over the training subjects, after G-SMOTE augmentation on PD-252, as $\gamma = [2 \cdot \text{median}(\delta^2) + 10^{-12}]^{-1}$. Rank $r=3$, fusion weight $w=0.5$, and SVM penalty $C=1$ were fixed across datasets; no per-dataset hyperparameter search was performed, thereby avoiding reuse of the outer test folds for hyperparameter selection [23,24]. The same protocol was applied to both cohorts. Within each outer training fold, resampling was governed solely by the observed training-fold class counts: if the counts were unequal, the minority class was raised to the majority count using G-SMOTE for the kernel models and ordinary SMOTE for the Euclidean comparators; if the counts were equal, resampling was the identity operation and no synthetic subjects were added. All UCI-489 training folds were balanced, so no synthetic subjects were generated there, whereas every PD-252 training fold activated the branch. The comparator set was identical across cohorts: the fused model; a matched Euclidean RBF SVM with fusion weight $w=0$, which shares the summary features, the resampling, and the kernel family of the fused model and differs from it only in the Grassmann kernel term (Figure 5B); a Grassmann-only kernel SVM with $w=1$; class-weighted logistic regression; and class-weighted linear SVM. On PD-252, where resampling is active, SMOTE plus logistic regression and SMOTE plus Euclidean RBF SVM were added; on UCI-489 these coincide with their unresampled counterparts. All methods were implemented in Python, and the implementation is available in the project repository (see the Data Availability Statement). Table 3 summarizes all fixed protocol choices. All comparators were implemented in the released codebase and verified against their stated configurations; the term audited comparators refers to this set, as distinct from values reported by other studies.

Each comparison isolates one design decision. Class-weighted logistic regression and class-weighted linear SVM probe whether simple decision boundaries on the Euclidean summary already suffice; SMOTE plus logistic regression tests a standard imbalanced-learning remedy [44]. The matched Euclidean RBF SVM is the critical ablation for the kernel term: any advantage of the fused model over it is attributable to the geometric kernel component, conditional on the shared augmentation, which itself selects partners by chordal distance (Section 2.7). SMOTE plus Euclidean RBF SVM is the fully Euclidean kernel pipeline; comparing it with the matched ablation isolates the geodesic augmentation, and comparing it with the fused model measures the complete geometric pipeline. On PD-252, G-SMOTE randomly samples anchor subjects and neighbors and draws interpolation coefficients, so performance varies with the augmentation seed. To isolate this source of variability, the fixed five-fold evaluation was repeated for 24 distinct augmentation seeds while the outer-fold assignments, preprocessing, hyperparameters, and all non-augmentation settings were held fixed. Within each seed and fold, the fused model and the matched Euclidean RBF ablation used the same augmented training set, so their difference is paired at the seed level. All PD-252 results of the original fused model use the reference implementation’s default augmentation seed, which is archived and regenerable; the seed

analysis of Section 3.3 characterizes variability around it. The adaptive-subspace fused model generates no synthetic subjects on either cohort and is therefore deterministic given the outer partition. The fused model, its matched Euclidean RBF ablation (which retains the G-SMOTE branch and is therefore not a fully Euclidean pipeline), class-weighted logistic regression, class-weighted linear SVM, and SMOTE plus logistic regression were refitted under rank normalization on the same folds, so that Table 4 compares these models under one normalization, while the original fused model and its matched ablation under z-standardization are retained as reference rows; the Grassmann-only kernel and the fully Euclidean SMOTE plus Euclidean RBF pipeline were evaluated under z-standardization only.

Table 3. Fixed experimental protocol. All learned components are fitted inside the outer training folds only.

Component	Setting
Outer evaluation	five-fold stratified subject-level cross-validation, shuffled, seed 42
Normalization	adaptive model: Gaussian rank normalization fitted on training recordings ($\min(300, n_k^{\text{rec}})$ quantile knots); original configuration: per-coordinate mean/scale from training recordings, zero-variance coordinates assigned unit scale
Supervised subspace	PLS projection fitted on training recordings with class-balanced targets; members $q \in \{8, 16, 32\}$ averaged after division by the training-fold decision SD; activated only when d exceeds the number of training recordings (PD-252: $753 > 603$; UCI-489: $45 < 192$); no synthetic subjects in this branch
Subject representation	rank $r = 3$ thin SVD (Grassmann view) + mean/dispersion vector (Euclidean view)
Kernel coefficients	$\gamma = [2 \text{ median}(\delta^2) + 10^{-12}]^{-1}$ from positive squared pairwise training distances (augmented training set on PD-252)
Fusion weight / penalty	$w = 0.5$; class-weighted SVM with $C = 1$; no per-dataset tuning
Resampling rule	training-fold class-count gate: identity when balanced (all UCI-489 folds); G-SMOTE to class parity when imbalanced (full-space branch; all PD-252 folds), five nearest minority neighbors under chordal distance; the supervised-subspace branch uses class weighting only
Comparators	adaptive-subspace fused model; fused and matched Euclidean RBF kernel SVMs (fusion weights 0.5 and 0) under rank normalization and under z-standardization; class-weighted logistic regression; class-weighted linear SVM; SMOTE + logistic regression where resampling is active; Grassmann-only kernel SVM ($w = 1$) and SMOTE + Euclidean RBF under z-standardization only
Metrics	balanced accuracy and macro-F1, mean $\pm$ fold-level SE; pooled out-of-fold confusion counts
Sensitivity analysis	20 outer subject partitions (all models, both normalizations); rank $r = 2$; 24 augmentation seeds; held-out permutation of Euclidean summary blocks (20 draws per family and fold), no refitting; subspace dimension sweep, forced gate, view ablation ($w = 0, w = 1$) and G-SMOTE inside the subspace branch

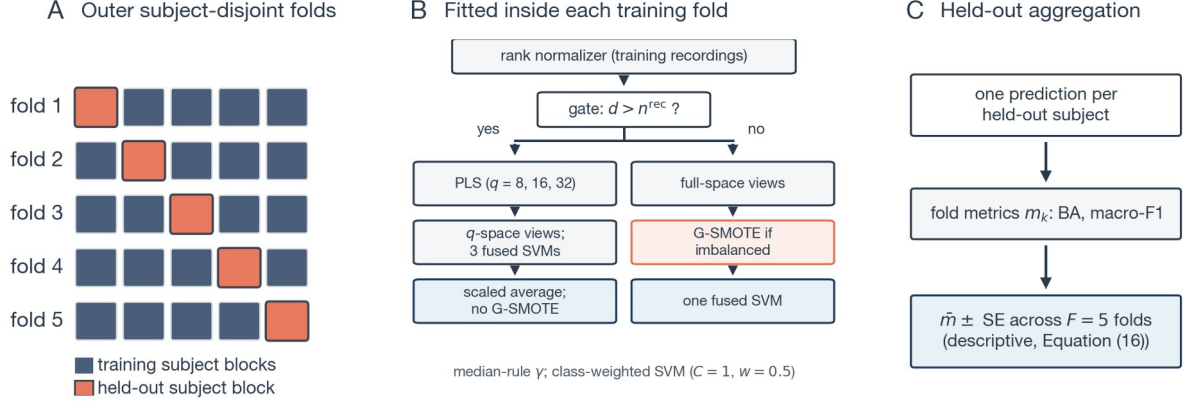


Figure 5. Evaluation protocol. (A) Five-fold subject-disjoint partition; in each outer fold one subject block (orange) is held out and never influences any fitted component. (B) Everything that learns from data is fitted inside the training fold only: the rank normalizer, then either the supervised-subspace branch (dimensionality gate active: PLS projections, q -space views, three SVMs averaged after scaling; no synthetic subjects) or the full-space branch (gate inactive: full-space views, conditional G-SMOTE in imbalanced folds, one SVM), with median-rule kernel coefficients in both branches. (C) Exactly one prediction is issued per held-out subject, fold metrics are computed at the subject level, and Equation (16) aggregates them descriptively.

2.12. Feature-Family Permutation Sensitivity

For feature family g , the fold-level decrease in performance and its summary are

$$I_{gkb} = m_k - m_{kb}^{\pi_g}, \hat{I}_{gk} = \frac{1}{B} \sum_{b=1}^B I_{gkb}, \hat{I}_g = \frac{1}{F} \sum_{k=1}^F \hat{I}_{gk},$$

$$\text{SE}(\hat{I}_g) = \left[\frac{1}{F(F-1)} \sum_{k=1}^F (\hat{I}_{gk} - \hat{I}_g)^2 \right]^{1/2}.$$

Sensitivity of the Euclidean summary was evaluated on each held-out fold without refitting. The coordinates of s_i associated with family g were jointly permuted across held-out subjects while Q_i and all other summary coordinates were held fixed. In Equation (22), m_k is the original held-out macro-F1 and $m_{kb}^{\pi_g}$ is macro-F1 after the b -th permutation draw. $B=20$ permutation draws were made per family and fold with a fixed random seed, the decrease was averaged over draws before the fold-level summary, and the six families follow the descriptor taxonomy of the source dataset (column assignments are archived with the code).

2.13. Model Development and Internal Robustness Analyses

Five groups of robustness analyses were specified during iterative development; none was preregistered. First, the entire five-fold evaluation was repeated for 20 alternative stratified subject partitions (StratifiedKFold seeds 0–19) with the default augmentation rule, and paired differences between models were summarized across partitions. Second, the fused model was rerun with rank $r=2$, for which every subject matrix has full rank, to test sensitivity to the rank choice and to the two rank-deficient subjects noted in Section 2.3. Third, on PD-252 the augmentation family was varied across the same 24 seeds as Section 3.3, so that the fused model, the matched Euclidean RBF ablation with G-SMOTE, and the fully Euclidean SMOTE plus Euclidean RBF pipeline were compared with paired draws. Fourth, the fusion weight was varied over $w \in \{0, 0.1, \dots, 0.8\}$ in every seed and partition run, and a nested variant selected w inside each training fold by repeated (three times five-fold) inner cross-validation on balanced accuracy over the same grid, so that the held-out folds were never used for selection. Exploratory alternatives (dimension-reduced, rank-one, centered, and block-balanced Grassmann views; decision-level fusion; a weight ensemble; other SVM penalties) are archived with the code. Fifth, for the adaptive-subspace fused model, the 20 outer partitions were rerun with the comparators of Table 4 under both normalizations (for models with the G-SMOTE gate, the augmentation seed is derived from the partition seed, so partition and augmentation realization vary

jointly; the archived fixed-augmentation partition results of the original model are used as a check in Section 3.5), the dimensionality gate was forced on for UCI-489 so that the subspace stage could be evaluated where the rule deactivates it, single members with $q \in \{4, 8, 16, 32, 64\}$ were evaluated against the three-member ensemble, the fusion weight was set to $w=0$ and $w=1$ inside the subspace ensemble to isolate the two views, and geodesic oversampling was applied inside the subspace branch across the 24 augmentation seeds of Section 3.3. The two adaptive components were designed after inspecting default-partition (seed 42) results of both cohorts and were then fixed before the partition analysis. The seed-42 estimates of the adaptive model are therefore development estimates, and the 20-partition results are an internal resampling-sensitivity analysis on the same 80 and 252 subjects; neither constitutes independent validation, partition-level counts are descriptive, and the repeated use of the same subjects for development and sensitivity analysis leaves an analysis-level selection optimism that only a new cohort can remove. All robustness analyses used the fixed hyperparameters of Table 3 unless stated and are archived with the reference implementation.

3. Results

3.1. Performance Across Datasets

Figure 6 and Table 4 report the comparators implemented under both normalizations at the default partition. On UCI-489 the dimensionality gate is inactive, so the adaptive-subspace fused model coincides with the rank-normalized fused model; it reached a balanced accuracy of 0.875 (SE 0.040) and a macro-F1 of 0.873 (SE 0.040), compared with 0.850 (SE 0.042) and 0.847 (SE 0.043) for the original z-standardized fused model and 0.863 and 0.860 for the rank-normalized Euclidean RBF ablation. Both kernel models exceeded rank-normalized class-weighted logistic regression (0.800 and 0.796) and linear SVM (0.800 and 0.797). On PD-252 the adaptive-subspace fused model reached 0.816 (SE 0.040) and 0.811 (SE 0.034). Rank normalization alone raised the full-space fused model from 0.731 and 0.749 (original configuration) to 0.782 and 0.781, and the supervised-subspace stage added a further +0.034 and +0.029; the three ensemble members individually reached 0.809, 0.814, and 0.798 balanced accuracy for $q=8, 16$, and 32 . The rank-normalized Euclidean RBF ablation reached 0.766 and 0.757, rank-normalized class-weighted logistic regression 0.769 and 0.774, class-weighted linear SVM 0.734 and 0.747, and SMOTE plus logistic regression 0.777 and 0.786. Under z-standardization the Grassmann-only kernel was insufficient on its own on both cohorts (balanced accuracy 0.600 on UCI-489 and 0.636 on PD-252) and the fully Euclidean SMOTE plus Euclidean RBF pipeline reached 0.728 and 0.714 on PD-252 (both archived with the code). These are descriptive cross-validation estimates at one outer partition, and for the adaptive model they are development estimates (Section 2.13); Section 3.5 reports the partition-sensitivity analysis, and no inferential test of model superiority was performed.

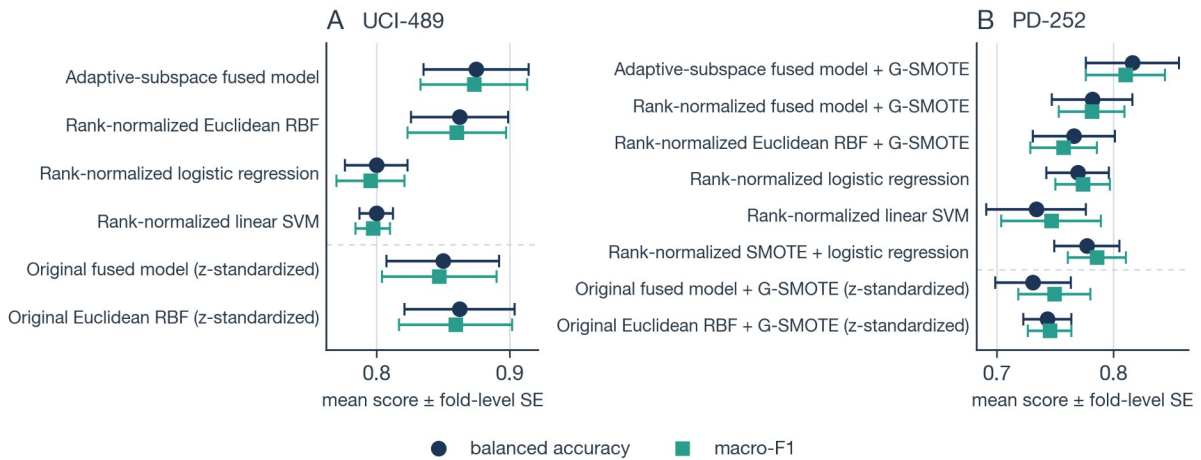


Figure 6. Subject-level scores of the comparators implemented under both normalizations at the default partition for (A) class-balanced UCI-489 and (B) high-dimensional, class-imbalanced PD-252, drawn from the archived out-

of-fold results. Rank-normalized models (upper rows) share one training-fold normalizer; the two lowest rows show the original z-standardized fused model and its matched ablation (PD-252 at the archived default augmentation seed); on PD-252 the Euclidean RBF rows retain the G-SMOTE branch. Markers denote mean balanced accuracy (circles) and macro-F1 (squares); whiskers denote fold-level standard error across five outer folds. On UCI-489 the adaptive-subspace model coincides with the rank-normalized fused model, and the SMOTE-based comparators coincide with their unresampled counterparts and are not shown.

Table 4. Subject-level performance at the default partition (mean $\pm$ fold-level standard error over five subject-disjoint folds; z-standardized PD-252 models at the archived default augmentation seed). All rank-normalized models share the same outer folds and training-fold normalizer; kernel models use the median-bandwidth rule and $C = 1$, whereas the linear comparators retain their model-specific fitting procedures. The Euclidean RBF rows on PD-252 retain the G-SMOTE branch. The adaptive-subspace fused model activates its supervised-subspace stage on PD-252 only. On UCI-489 the SMOTE-based comparators coincide with their unresampled counterparts and are omitted. The seed-42 estimates of the adaptive model are development estimates (Section 2.13); Table 7 reports the partition-sensitivity analysis.

Dataset	Model	Balanced acc.	Macro-F1
UCI-489	Adaptive-subspace fused model (gate inactive: rank-normalized fused model)	0.875 ± 0.040	0.873 ± 0.040
UCI-489	Rank-normalized Euclidean RBF (matched ablation)	0.863 ± 0.036	0.860 ± 0.037
UCI-489	Rank-normalized class-weighted logistic regression	0.800 ± 0.023	0.796 ± 0.026
UCI-489	Rank-normalized class-weighted linear SVM	0.800 ± 0.012	0.797 ± 0.013
UCI-489	Original fused model (z-standardized)	0.850 ± 0.042	0.847 ± 0.043
UCI-489	Original Euclidean RBF (z-standardized)	0.863 ± 0.041	0.860 ± 0.042
PD-252	Adaptive-subspace fused model	0.816 ± 0.040	0.811 ± 0.034
PD-252	Rank-normalized fused model + G-SMOTE (full space)	0.782 ± 0.035	0.781 ± 0.028
PD-252	Rank-normalized Euclidean RBF + G-SMOTE	0.766 ± 0.035	0.757 ± 0.028
PD-252	Rank-normalized class-weighted logistic regression	0.769 ± 0.027	0.774 ± 0.023
PD-252	Rank-normalized class-weighted linear SVM	0.734 ± 0.043	0.747 ± 0.043
PD-252	Rank-normalized SMOTE + logistic regression	0.777 ± 0.028	0.786 ± 0.025
PD-252	Original fused model (z-standardized, G-SMOTE)	0.731 ± 0.032	0.749 ± 0.031
PD-252	Original Euclidean RBF (z-standardized, G-SMOTE)	0.743 ± 0.021	0.745 ± 0.019

3.2. Operating Characteristics

The UCI-489 adaptive-subspace fused model produced 36 true-negative (control), 4 false-positive, 6 false-negative, and 34 true-positive (PD) classifications, yielding sensitivity 0.850, specificity 0.900, positive predictive value (PPV) 0.895, and negative predictive value (NPV) 0.857. On PD-252 the pooled confusion matrix contained 169 true-positive, 19 false-negative, 47 true-negative, and 17 false-positive classifications, corresponding to sensitivity 0.899, specificity 0.734, PPV 0.909, and NPV 0.712 (Figure 7, Table 5); the original fused model at the archived default augmentation seed had sensitivity 0.931 and specificity 0.531 (175, 13, 34, and 30 classifications), so the adaptive model traded a small loss of sensitivity for a substantially higher specificity. Sensitivity and specificity do not depend algebraically on prevalence but may change under spectrum and setting shifts, whereas PPV and NPV additionally depend directly on this cohort’s 188/64 composition and the default decision threshold and do not transport to lower-prevalence screening settings. All values in this section are development-partition estimates. Rates computed from pooled confusion matrices are descriptive and need not equal the arithmetic mean of fold-level metrics. Even at the improved operating point, roughly one control in four would be referred at the default threshold, which implies a substantial false-referral burden in lower-

prevalence screening populations. This operating profile warrants further evaluation of the method as a research-stage screening aid and does not support its use for autonomous diagnosis.

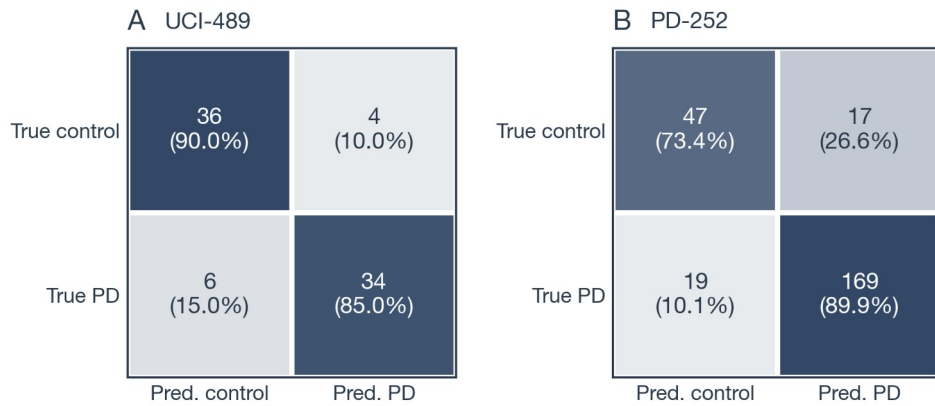


Figure 7. Pooled out-of-fold confusion matrices of the adaptive-subspace fused model for (A) UCI-489 and (B) PD-252 at the default partition, containing one prediction per subject; percentages are normalized within the true class. The model generates no synthetic subjects, so the counts do not depend on an augmentation seed. These are seed-42 development-partition estimates and are not externally validated operating characteristics.

Table 5. Pooled out-of-fold operating characteristics of the adaptive-subspace fused model computed from the confusion counts of Figure 7 (UCI-489: TP = 34, FN = 6, TN = 36, FP = 4; PD-252: TP = 169, FN = 19, TN = 47, FP = 17). TP, true positive; FN, false negative; TN, true negative; FP, false positive; PPV, positive predictive value; NPV, negative predictive value.

Dataset	Sensitivity	Specificity	PPV	NPV
UCI-489	0.850	0.900	0.895	0.857
PD-252	0.899	0.734	0.909	0.712

3.3. Sensitivity to the G-SMOTE Augmentation Seed

The PD-252 scores of the z-standardized models in Table 4 were obtained at the archived default augmentation seed; the adaptive-subspace fused model generates no synthetic subjects and is unaffected by this seed. The paired analysis of Section 2.11 characterizes the sensitivity of both models and of their difference to the augmentation seed (Table 6). Across 24 seed runs, balanced accuracy was 0.739 ± 0.012 for the fused model and 0.731 ± 0.013 for the matched Euclidean ablation, and macro-F1 was 0.752 ± 0.013 and 0.729 ± 0.013 , respectively (mean $\pm$ standard deviation across seeds). The paired macro-F1 difference averaged $+0.023 \pm 0.015$ and was positive in 23 of 24 runs. The paired balanced-accuracy difference averaged $+0.009 \pm 0.014$, was positive in 16 of 24 runs, and reversed sign in 8 of 24. The archived default-seed realization of Tables 4 and 5 (0.731 and 0.749) lies within the seed ranges [0.722, 0.760] and [0.732, 0.777]. Across the same 24 seeds, the fully Euclidean SMOTE plus Euclidean RBF pipeline averaged 0.726 ± 0.009 and 0.718 ± 0.010 ; replacing ordinary SMOTE by G-SMOTE in the Euclidean RBF model changed balanced accuracy by $+0.005 \pm 0.016$ and macro-F1 by $+0.011 \pm 0.016$ (positive in 15 and 17 of 24 seeds), and the complete geometric pipeline exceeded the fully Euclidean pipeline by $+0.014 \pm 0.015$ and $+0.034 \pm 0.016$ (positive in 19 and 24 of 24 seeds; Table 7). Thus, conditional on the fixed outer folds, the macro-F1 margin of the original fused model over its ablation was directionally consistent in all but one seed run, whereas the balanced-accuracy margin can be reversed by individual draws. The corresponding analysis of geodesic oversampling inside the supervised subspace is reported in Section 3.5.

Table 6. Sensitivity of the PD-252 comparison to the G-SMOTE augmentation seed (original z-standardized configuration): paired evaluation of the fused model and the matched Euclidean RBF ablation across 24 distinct seeds with outer folds, preprocessing, and hyperparameters held fixed. Values are means $\pm$ standard deviations (SD) across seeds; Δ denotes the paired difference (fused minus ablation) computed within each seed, summarized by its mean $\pm$ SD, median [interquartile range, IQR], and sign count.

Metric	Fused	Ablation	Δ mean $\pm$ SD	Δ median [IQR]	$\Delta > 0$
Balanced accuracy	0.739 ± 0.012	0.731 ± 0.013	$+0.009 \pm 0.014$	$+0.008 [-0.002, +0.018]$	16/24
Macro-F1	0.752 ± 0.013	0.729 ± 0.013	$+0.023 \pm 0.015$	$+0.020 [+0.012, +0.034]$	23/24

Each seed-level value is the five-fold cross-validation point estimate under fixed outer folds; standard deviations describe variation across the 24 augmentation seeds and are distinct from the fold-level standard errors of Table 4. Δ ranges: balanced accuracy $[-0.013, +0.042]$; macro-F1 $[-0.001, +0.061]$. Per-seed values are archived in the project repository.

3.4. Feature-Family Sensitivity

For the original z-standardized fused model, the TQWT summary block produced the largest mean decrease in held-out macro-F1, at 0.149 with an SE of 0.020, followed by the MFCC summary block, at 0.046 with an SE of 0.008 (Figure 8); within a fold, the standard deviation across the 20 permutation draws averaged 0.067 for TQWT. The vocal/time-frequency, wavelet, IMF/EMD, and demographic families produced mean decreases near zero. These exploratory values measure dependence of the fitted model on the Euclidean summary coordinates under correlated features; they do not establish biological causality and, because only the Euclidean view is permuted, do not quantify the dependence of the full two-view model on each family. The prominence of the TQWT family is consistent with the descriptor engineering of the source dataset, where tunable Q-factor features were introduced precisely for Parkinsonian voice [31,55].

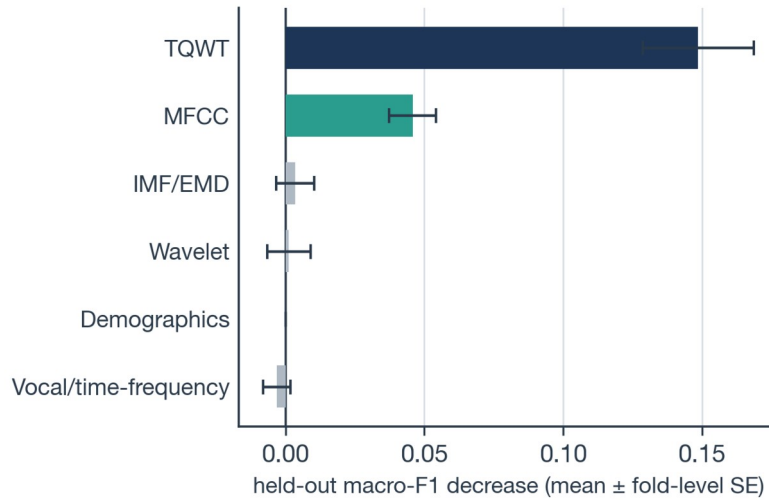


Figure 8. PD-252 feature-family permutation sensitivity of the fused model on the Euclidean summary view under the archived default-seed realization (20 permutation draws per family and fold). Positive values denote a decrease in held-out macro-F1 after permutation, and whiskers denote fold-level standard error. TQWT denotes tunable Q-factor wavelet transform; MFCC denotes Mel-frequency cepstral coefficients; IMF/EMD denotes intrinsic mode function/empirical mode decomposition; Demographics denotes the dataset-provided sex/gender covariate.

3.5. Robustness to the Outer Partition, the Subspace Dimension, the Rank, and the Augmentation Family

Figure 9 and Table 7 summarize the internal partition-sensitivity analysis of Section 2.13. Across 20 alternative outer subject partitions on PD-252, the adaptive-subspace fused model averaged 0.784 ± 0.017 balanced accuracy and 0.785 ± 0.014 macro-F1 (mean $\pm$ SD across partitions), compared with 0.721 ± 0.020 and 0.734 ± 0.019 for the original fused model; the paired differences were $+0.062 \pm 0.018$ and $+0.050 \pm 0.017$, positive in all 20 partitions on both metrics; because the original model’s augmentation

seed varies with the partition in these runs, the comparison was repeated against the archived original-model partition results with a fixed augmentation seed, giving $+0.059 \pm 0.016$ and $+0.048 \pm 0.017$, again positive in all 20 partitions. Rank normalization alone accounted for $+0.028$ and $+0.017$ of this margin (rank-normalized fused model versus original fused model, positive in 19 and 17 of 20 partitions), and the supervised-subspace stage for the remainder (adaptive-subspace versus rank-normalized fused model: $+0.034$ and $+0.033$, positive in 19 and 20 of 20 partitions). The adaptive model also exceeded rank-normalized class-weighted logistic regression in 19 of 20 partitions on both metrics. On UCI-489, where the gate is inactive, the adaptive model coincides with the rank-normalized fused model and had similar mean estimates to the original fused model (0.853 ± 0.019 versus 0.853 ± 0.013 balanced accuracy; paired difference $+0.001 \pm 0.021$, positive in 9 of 20 partitions with 4 ties; macro-F1 $+0.001 \pm 0.022$, positive in 11 of 20). Forcing the subspace stage on for UCI-489 lowered balanced accuracy by 0.011 on average (higher in only 4 of 20 partitions, with 4 ties), which is consistent with the gate rule, although the mechanism was not tested directly. Partition-to-partition standard deviations of the adaptive model (0.017 and 0.014 on PD-252) were of the same order as those of the original model (0.020 and 0.019).

Figure 9B shows the subspace-dimension sweep with the gate forced on for both cohorts. On PD-252 single members reached 0.769 ($q=4$), 0.772 ($q=8$), 0.782 ($q=16$), 0.782 ($q=32$), and 0.775 ($q=64$) mean balanced accuracy across partitions, and the three-member ensemble 0.784, so the fixed-dimension average was at least as good as any single dimension without using held-out subjects to choose one; on UCI-489 every subspace dimension (0.839–0.843) fell below the full-space model (0.853). Inside the subspace branch, geodesic oversampling to class parity changed balanced accuracy by -0.019 ± 0.008 and macro-F1 by -0.007 ± 0.008 across the 24 augmentation seeds (positive in 1/24 and 7/24 seeds; Table 7), so the class-weighted SVM without synthetic subjects was retained for this branch. Setting the fusion weight to $w=1$ inside the subspace ensemble (Grassmann view only) gave 0.636 balanced accuracy on PD-252 and 0.579 on UCI-489 across partitions, and $w=0$ (Euclidean view only) gave 0.779 and 0.851, so the fused ensemble at $w=0.5$ exceeded the Euclidean-only ensemble by $+0.005$ on PD-252 (positive in 12 of 20 partitions) and by $+0.002$ on UCI-489 (9 of 20, 2 ties): the Euclidean view therefore carried most of the adaptive model’s predictive performance; adding the Grassmann term produced small, partition-inconsistent mean increments, and the Grassmann-only ensemble performed poorly.

The robustness analyses of the original z-standardized configuration are summarized here and archived in full. Across the 20 partitions, the original fused model and its matched Euclidean RBF ablation had similar mean estimates on both cohorts (paired balanced-accuracy differences $+0.004 \pm 0.013$ on UCI-489 and -0.000 ± 0.016 on PD-252; macro-F1 $+0.005 \pm 0.013$ and $+0.008 \pm 0.017$), whereas the fused model retained a macro-F1 advantage of $+0.028 \pm 0.019$ over the fully Euclidean SMOTE plus Euclidean RBF pipeline on PD-252 (18 of 20 partitions). Reducing the rank to $r=2$ changed the original fused model by less than one fold-level standard error (UCI-489: 0.838 and 0.834; PD-252: 0.739 and 0.755); this analysis concerns the original configuration only and was not repeated for the adaptive model. Figure 10 shows the sensitivity of the original model to the fusion weight: across the 24 PD-252 augmentation seeds, weights $w=0.2$ – 0.7 exceeded the $w=0$ ablation in macro-F1 in 22 or 23 runs, whereas $w=0.1$ did so in 15 runs; the highest mean balanced accuracy occurred at $w=0.3$ and the highest mean macro-F1 at $w=0.7$, and on UCI-489 the highest observed means also occurred at larger weights ($w=0.7$); a nested selection of the weight inside each training fold made the balanced-accuracy improvement of the original model consistent (20 of 24 seeds, 13 of 20 partitions) at a small magnitude, and none of the archived exploratory variants of the original configuration (dimension-reduced, rank-one, centered, and block-balanced Grassmann views, decision-level fusion, weight ensembles, other penalties) exceeded it robustly. The comparisons of Table 7 are paired within partitions on identical folds.

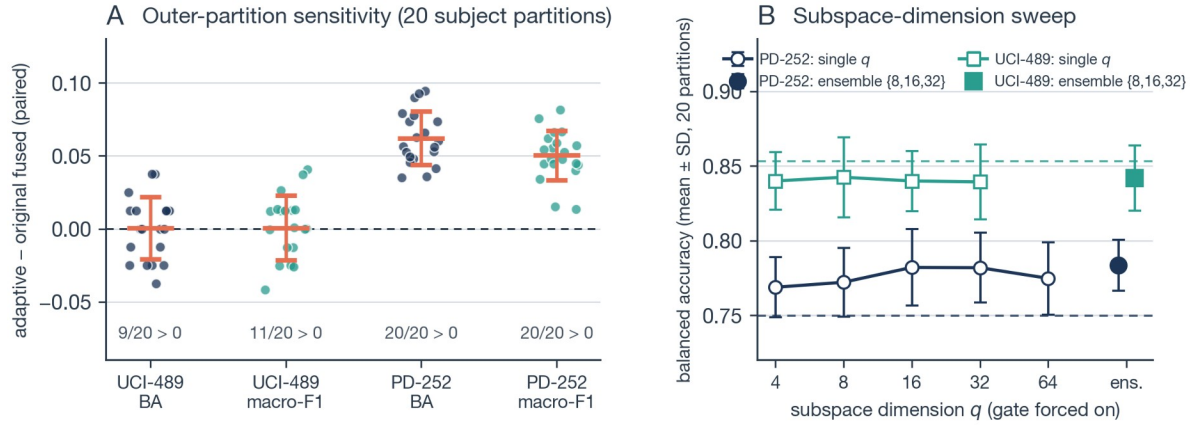


Figure 9. Internal robustness of the adaptive-subspace fused model. (A) Paired differences between the adaptive-subspace fused model and the original z-standardized fused model across 20 alternative outer subject partitions (points; coral bars denote mean $\pm$ SD; counts give the number of partitions with a positive difference), for balanced accuracy (BA, navy) and macro-F1 (teal) on both cohorts; the original model’s augmentation seed varies jointly with the partition, and on UCI-489 the gate is inactive so the adaptive model coincides with the rank-normalized fused model. (B) Subspace-dimension sweep with the gate forced on: mean balanced accuracy across the 20 partitions ($\pm$ SD) for single members $q \in \{4, 8, 16, 32, 64\}$ ($q \leq 32$ on UCI-489), the three-member ensemble (filled markers), and the full-space rank-normalized fused model (dashed line), for both cohorts.

Table 7. Internal robustness analyses of the adaptive-subspace fused model (Section 2.13). Each row pairs two models within the same outer subject partition (20 partitions, StratifiedKFold seeds 0–19; for models with the G-SMOTE gate the augmentation seed is derived from the partition seed, so partition and augmentation realization vary jointly) or the same augmentation seed (24 seeds at the default partition); values are means $\pm$ standard deviations across partitions or seeds, Δ is the paired difference (model A minus model B), and the last column counts strictly positive differences (ties noted where they occur). On UCI-489 the dimensionality gate is inactive, so the adaptive-subspace model coincides with the rank-normalized fused model; the forced-gate row applies the subspace stage there regardless. The subspace branch generates no synthetic subjects, so model B of the G-SMOTE ablation is fixed and only model A varies with the augmentation seed.

Analysis (A vs. B)	Cohort	Metric	Model A	Model B	Δ (A – B)	$\Delta > 0$
Outer partition (20): adaptive-subspace vs. original fused	UCI-489	BA	0.853 ± 0.019	0.853 ± 0.013	$+0.001 \pm 0.021$	9/20 (4 ties)
	UCI-489	macro-F1	0.852 ± 0.019	0.851 ± 0.014	$+0.001 \pm 0.022$	11/20
	PD-252	BA	0.784 ± 0.017	0.721 ± 0.020	$+0.062 \pm 0.018$	20/20
	PD-252	macro-F1	0.785 ± 0.014	0.734 ± 0.019	$+0.050 \pm 0.017$	20/20
Outer partition (20): adaptive-subspace vs. rank-normalized fused	PD-252	BA	0.784 ± 0.017	0.750 ± 0.016	$+0.034 \pm 0.017$	19/20
	PD-252	macro-F1	0.785 ± 0.014	0.751 ± 0.018	$+0.033 \pm 0.017$	20/20
Outer partition (20): rank- normalized fused vs. original fused	UCI-489	BA	0.853 ± 0.019	0.853 ± 0.013	$+0.001 \pm 0.021$	9/20 (4 ties)
	UCI-489	macro-F1	0.852 ± 0.019	0.851 ± 0.014	$+0.001 \pm 0.022$	11/20
	PD-252	BA	0.750 ± 0.016	0.721 ± 0.020	$+0.028 \pm 0.023$	19/20
	PD-252	macro-F1	0.751 ± 0.018	0.734 ± 0.019	$+0.017 \pm 0.023$	17/20
Outer partition (20): adaptive-subspace vs. rank-normalized logistic regression	UCI-489	BA	0.853 ± 0.019	0.828 ± 0.027	$+0.025 \pm 0.030$	16/20 (1 tie)
	UCI-489	macro-F1	0.852 ± 0.019	0.826 ± 0.027	$+0.026 \pm 0.030$	17/20
	PD-252	BA	0.784 ± 0.017	0.752 ± 0.024	$+0.032 \pm 0.020$	19/20
	PD-252	macro-F1	0.785 ± 0.014	0.757 ± 0.024	$+0.028 \pm 0.018$	19/20

Outer partition (20): subspace stage forced on vs. gate rule (rank- normalized fused)	UCI-489	BA	0.842 ± 0.022	0.853 ± 0.019	-0.011 ± 0.021	4/20 (4 ties)
	UCI-489	macro-F1	0.840 ± 0.022	0.852 ± 0.019	-0.011 ± 0.022	6/20 (1 tie)
Augmentation seed (24): subspace branch with G- SMOTE vs. without	PD-252	BA	0.797 ± 0.008	0.816 (fixed)	-0.019 ± 0.008	1/24
	PD-252	macro-F1	0.804 ± 0.008	0.811 (fixed)	-0.007 ± 0.008	7/24

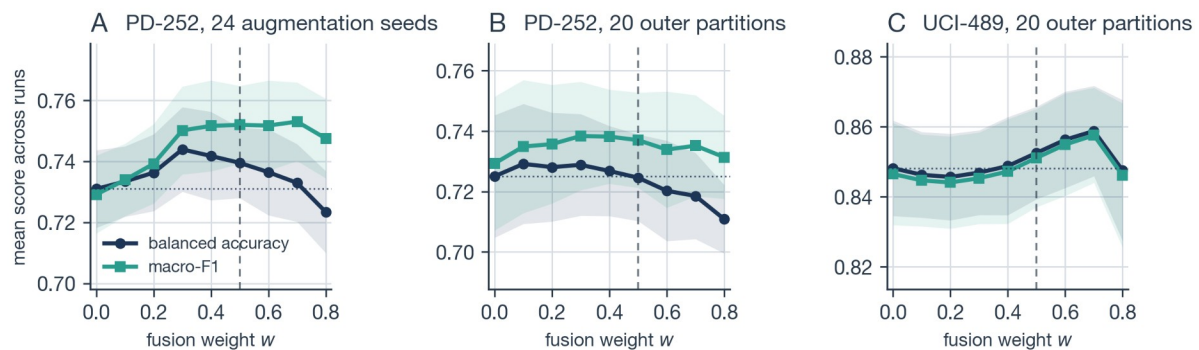


Figure 10. Sensitivity of the original z-standardized fused model to the fusion weight w under the fixed protocol: mean balanced accuracy (navy circles) and macro-F1 (teal squares) across (A) the 24 augmentation seeds on PD-252, (B) the 20 outer partitions on PD-252, and (C) the 20 outer partitions on UCI-489. Shaded bands denote ± 1 standard deviation across runs, the dashed line marks the prespecified weight $w=0.5$, and the dotted line marks the balanced accuracy of the matched Euclidean RBF ablation ($w=0$).

4. Discussion

The central methodological contribution is the alignment of the evaluation unit with the clinical unit. Subject-disjoint validation asks whether a model generalizes to an unseen person. The Grassmann representation uses the dependence among repeated recordings rather than treating the recordings as independent replicates. Basis invariance removes arbitrary coordinate choices, while the Euclidean view restores the coordinate-level location and dispersion magnitudes that a pure subspace representation discards.

Three findings follow from the results. First, the original fused model improved macro-F1 over the matched Euclidean RBF ablation in 23 of 24 augmentation seeds and 13 of 20 outer partitions on PD-252, but its balanced-accuracy advantage at the fixed weight vanished across partitions; the optimal weight was cohort-dependent, and the geometric machinery paid off mainly when geodesic augmentation and the geometric kernel were combined (macro-F1 above a fully Euclidean kernel pipeline in 24 of 24 seeds and 18 of 20 partitions). Second, two fold-local remedies for properties of the descriptors themselves produced the largest and most consistent gains: rank normalization was associated with higher distance variability and higher performance on PD-252, consistent with a reduced influence of extreme coordinates, and the supervised-subspace stage increased chordal-distance variability where descriptors and about 600 training recordings had driven the chordal distances toward their maximum; the view ablation nevertheless attributes most of the adaptive model's performance to the rank-normalized supervised Euclidean summary, with the Grassmann term adding small, partition-inconsistent increments. On PD-252 the adaptive-subspace fused model exceeded the original fused model in every one of 20 outer partitions on both metrics (+0.062 balanced accuracy, +0.050 macro-F1) and raised specificity at the default threshold from 0.531 to 0.734. Third, the dimensionality gate matters: on the balanced 45-dimensional UCI-489 cohort the same supervised stage lowered balanced accuracy, and the gate rule left the rank-normalized fused model in place, where its mean estimates were similar

to those of the original model. The gate rule ($d > n_k^{\text{rec}}$) is interpretable and was fixed before the partition analysis, but it was formulated after both cohorts had been inspected, so its generality is untested on new cohorts. None of these counts is population-level inference about model superiority. With respect to the objectives of Section 1, the first was addressed analytically: Equations (17) and (18) formalize the difference between record-wise and unseen-person risks. The second receives an affirmative but modest answer for the original configuration, whereas the third receives a large and partition-consistent answer on the high-dimensional cohort, driven mainly by the Euclidean view, and a null answer on the low-dimensional cohort. Whenever repeated recordings from previously observed people are easier to classify than recordings from unseen people, record-wise scores are optimistically biased in direct proportion to the probability that a validation recording belongs to an already-observed person. Subject-disjoint folds remove that term entirely. Reported subject-level scores are therefore expected to be lower than record-wise figures reported on the same corpora, but they answer the question that a screening deployment poses [20,22].

An evaluation-focused perspective also clarifies how these results should be interpreted in relation to the wider literature. Reported accuracies for voice-based PD detection vary widely across studies, and validation strategy is a recognized source of that variation [16,17]. Record-wise splitting remains common in secondary analyses of repeated-recording corpora, whereas replication-aware methods have also been developed for UCI-489 [29]. Balanced accuracy and macro-F1 were used because both cohorts were evaluated at the subject level and PD-252 had unequal class sizes. Both metrics weight the diagnostic classes symmetrically rather than rewarding majority-class predictions [64,65]. Fold-level standard errors are reported descriptively, and no significance test is claimed: five outer folds provide a descriptive summary of variability, not a basis for inferential comparison [25].

A separate research-stage web interface (Figure 11) demonstrates browser-based capture with a commodity microphone for three speech tasks (sustained vowel, read passage using an excerpt of the Rainbow Passage [69], and free speech), together with an illustrative longitudinal display. The application was not connected to either evaluated cohort and was not assessed for usability, calibration, or clinical performance. Figure 12 places the prototype in a proposed, rather than validated, deployment workflow and distinguishes the components evaluated here from capture and decision-support stages that were not; the evaluated models used only three repeated sustained-vowel recordings per subject, and constructing a common representation from heterogeneous application tasks remains future work. Such low-friction, repeated measurement is consistent with technology roadmaps for PD care [50,51]. The application is a research-stage prototype, not an autonomous diagnostic system or medical device, and no application-collected data were used in the analyses reported here. Prospective use requires explicit privacy, consent, security, accessibility conformance, calibration, and clinical governance procedures.

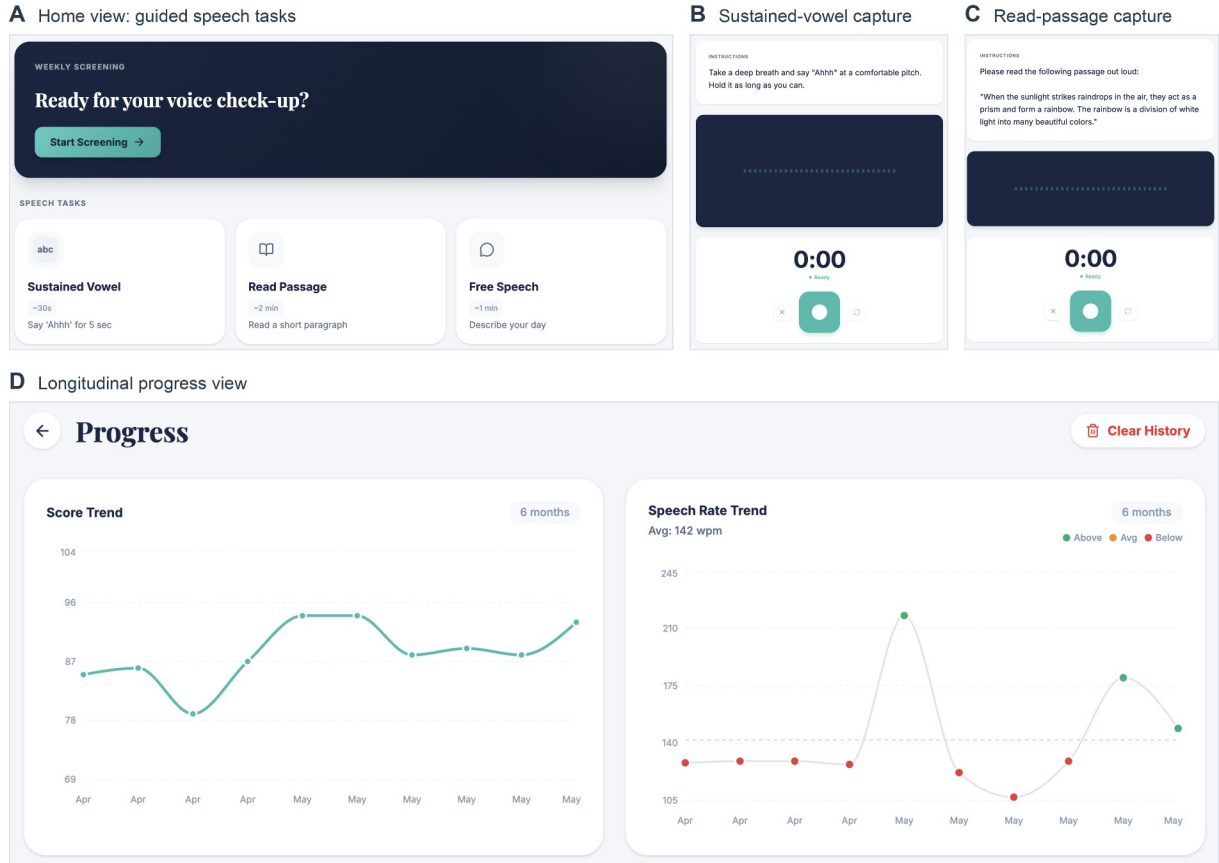


Figure 11. VoxNeuro web application (publicly accessible research-stage prototype). (A) Home view listing the three guided speech tasks. (B) Sustained-vowel capture screen. (C) Read-passages capture screen showing an excerpt from the Rainbow Passage. (D) Longitudinal progress view with illustrative entries. The application requires only a commodity microphone and a web browser; no application-collected data were used in the analyses reported here.

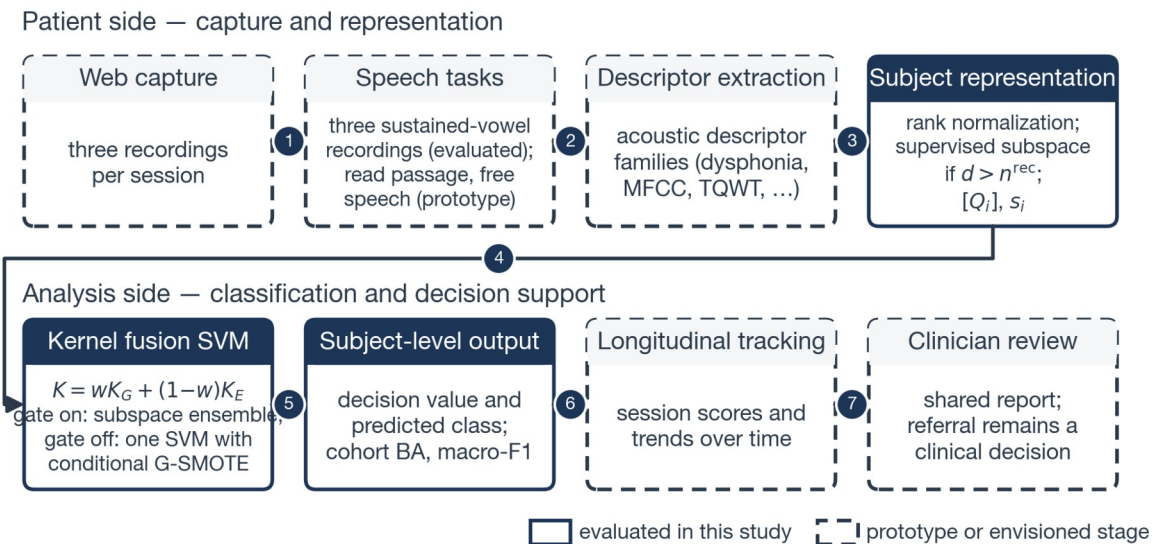


Figure 12. Study pipeline and envisioned deployment workflow of the VoxNeuro prototype. Solid boxes mark the components evaluated offline in this study on the two retrospective public cohorts (rank normalization and subject representation, with the supervised-subspace branch in the high-dimensional regime and the full-space branch with conditional G-SMOTE otherwise; kernel fusion; and subject-level scoring); dashed boxes mark capture and decision-support stages of the research-stage prototype that were not evaluated here, and no end-to-end pipeline from captured audio to classification was evaluated. Descriptor extraction in this study used the dataset-provided

engineered features, and any referral decision remains a clinical judgment. The evaluated models used only three repeated sustained-vowel recordings per subject; the read-passage and free-speech tasks are prototype capture tasks that were not used in any evaluated model.

This study has limitations. The speech cohorts contain retrospective engineered features rather than prospectively acquired raw audio. The study included no probability calibration [70,71], decision-curve analysis [72], independent clinical test cohort, device-shift evaluation, language-shift audit, or prespecified subgroup analysis. Sensitivity to the fusion weight, the outer partition, the augmentation seed, the augmentation family, the subspace rank, the subspace dimension, and the normalization was characterized, but sensitivity to the SVM penalty, neighbor count, and median rule for the kernel coefficients was not, and the gains attributable to the geometric kernel term alone are small relative to fold-level variability. The adaptive components were developed after inspecting default-partition results and were confirmed only by repeated cross-validation on the same 80 and 252 subjects; the seed-42 estimates of the adaptive model are development estimates, the 20-partition results are not independent replications, and the gate rule has been exercised on exactly two cohorts. The PLS projection treats the three correlated recordings of a subject as separate observations when it is fitted, and the number of quantile knots, the subspace dimensions, and the equal-weight ensemble were fixed by convention rather than optimized. The comparator set did not include deep models, and pretrained audio models could not be evaluated because raw audio was unavailable. The Grassmann view alone was insufficient in both cohorts (balanced accuracy 0.600 and 0.636 under z-standardization; 0.579 and 0.636 inside the adaptive model), and the rank-two sensitivity analysis concerns the original configuration only. Because high PD-252 sensitivity coexists with modest specificity even after the adaptive improvements, a credible system should support referral and longitudinal review rather than replace neurological assessment. Future work should preregister preprocessing and thresholds and report in accordance with structured guidance for clinical prediction models [73,74]; collect repeated raw speech prospectively; and evaluate calibration, test-retest reliability, fairness, and clinician-facing decision consequences. The two-view construction extends naturally to additional recording tasks and views through further positive-semidefinite kernel terms, with weights selected by nested multiple-kernel optimization [42,43]; an exploratory family-wise alignment-weighted kernel combination is archived with the code but was not adopted because it degraded the low-dimensional cohort. External validation of the frozen adaptive model on a cohort that played no role in its development is the decisive next step.

Reproducibility safeguards included fixed subject-disjoint folds, training-fold-only preprocessing, fixed hyperparameters, saved fold-level metrics, pooled confusion counts, and restriction of reported results to audited comparator implementations. The project repository archives, under a pinned dependency environment, subject-level out-of-fold predictions with fold identifiers for both cohorts, the unified comparator results under both normalizations, the 24-seed paired metrics, the partition, rank, augmentation-family, and permutation outputs with the feature-family column map, the adaptive-model package (partition analysis, dimension sweep, view and oversampling ablations, concentration diagnostics), and the scripts that generate them; the archived scripts and outputs permit regeneration and audit of the reported numerical summaries under the pinned software environment, and the adaptive model is deterministic given the partition and single-threaded linear algebra.

5. Conclusions

VoxNeuro provides a leakage-safe framework for repeated Parkinsonian speech. Each subject is represented by a Grassmann subspace and a Euclidean mean-dispersion vector, the corresponding positive-semidefinite kernels are fused, and every learned component, including Gaussian rank normalization and, in the high-dimensional regime, a supervised-subspace ensemble, is fitted inside the training fold. Under one fixed configuration the adaptive-subspace fused model reached balanced accuracies of 0.875 on UCI-489 and 0.816 on PD-252 at the development partition and, across 20 additional overlapping partitions of the same subjects, yielded higher PD-252 point estimates than the

original fused model in every partition (+0.062 balanced accuracy, +0.050 macro-F1) while yielding similar estimates on the balanced, low-dimensional UCI-489 cohort; most of the improvement is attributable to normalization and the supervised Euclidean summary rather than to the Grassmann term. A publicly accessible research-stage web prototype implements guided capture and longitudinal visualization. Clinically credible digital biomarkers require auditable implementations, transparent uncertainty, and evaluation designs that test generalization to unseen people; prospective external validation of the frozen model is the necessary next step before any clinical use.

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Institutional Review Board Statement: Ethical review and approval were not required for this study, as it consisted exclusively of a secondary computational analysis of publicly available, fully de-identified datasets; no participants were recruited, and no new human data were collected.

Informed Consent Statement: Not applicable. The study analyzed publicly available, de-identified datasets and did not involve direct human participation; informed consent and ethics procedures for the original data collections are documented in the source publications cited in Section 2.1.

Data Availability Statement: The datasets analyzed in this study are openly available from the UCI Machine Learning Repository: the Parkinson Dataset with Replicated Acoustic Features (UCI-489; UCI dataset 489, doi:10.24432/C5701F) and the Parkinson’s Disease Classification dataset (PD-252; UCI dataset 470, doi:10.24432/C5MS4X). Source code is available at <https://github.com/khan-laiba/voxneuro> (release v2.0.0); the repository also archives, under the pinned dependency environment, subject-level out-of-fold prediction files for both cohorts and both normalizations, the adaptive-model package (comparators, 20-partition analysis, subspace-dimension sweep, view and oversampling ablations, and concentration diagnostics), the unified comparator results, the per-seed outputs of the PD-252 seed-sensitivity and augmentation-family analyses, the outer-partition and rank sensitivity results, the fusion-weight sweep, the nested-selection and exploratory-variant outputs, and the permutation analysis with its feature-family column map. The publicly accessible research-stage web application is available at <https://voxneuro.org>. No new primary data were collected in this study.

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